International Migration Network Analysis
A Network Science and Machine Learning Project

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Abstract

This report delivers a structured, end-to-end analysis of global migration using network science and machine learning. Countries are modeled as nodes and migration flows as directed weighted edges across eight snapshots (1990–2024). The study combines dynamic network construction, centrality metrics, community evolution, temporal drift measures, and regression modeling. We introduce a custom Migration Influence Score (MIS) that blends PageRank with total flow volume to surface practical hubs. Outputs include tables, figures, and dashboard-ready artifacts to support interpretation and presentation.

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Executive Summary

- Built 8 dynamic migration networks (1990–2024) from UN DESA data.
- Identified global hubs using centrality and a custom MIS metric.
- Measured community stability using NMI, ARI, and Jaccard drift.
- Achieved strong predictive performance with regression models ($R^2 \approx 0.95$).
- Delivered dashboards and reproducible artifacts for reporting.

Chapter 1

Introduction

International migration reshapes demographics, economies, and policy. Network science provides a systems view of migration by modeling countries as nodes and flows as directed weighted edges. This report focuses on structural hubs, community evolution, and temporal stability, and extends the analysis with machine learning and a custom influence metric.

Chapter 2

Problem Statement and Objectives

2.1 Problem Statement

How can migration from 1990 to 2024 be modeled as a dynamic network to reveal hubs, community structure, and temporal drift, while linking flows to explanatory features?

2.2 Objectives

- Build a dynamic directed migration network for multiple snapshots.
- Compute centrality metrics to identify origin and destination hubs.
- Detect communities and quantify stability over time.
- Introduce a custom influence metric for practical ranking.
- Train regression models to evaluate predictive relationships.
- Produce reproducible outputs and dashboards.

Chapter 3

Dataset and Preprocessing

3.1 Data Sources

Primary: UN DESA International Migrant Stock 2024.
Auxiliary (when available): GDP, population, unemployment, conflict indicators.

3.2 Preprocessing Workflow

- Removed aggregate/non-country rows and standardized country identifiers.
- Converted matrix-form data to long-format origin–destination–year rows.
- Validated numeric ranges and handled missing values.

3.3 Snapshot Summary

Table 3.1: Snapshot Summary (Nodes, Edges, Total Migrants)

Year	Nodes	Edges	Total Migrants
1990	234	8235	153,916,063
1995	234	8468	163,176,002
2000	234	8593	174,566,152
2005	234	8727	192,788,721
2010	234	9013	221,020,392
2015	234	9264	250,042,020
2020	234	9269	275,284,032
2024	234	9273	304,021,813

Chapter 4

Exploratory Analysis

Migration flows are heavy-tailed: a small number of countries dominate inflow and outflow. The most intense corridors are highly concentrated, indicating structural hubs and persistent routes.

4.1 Top Destinations (2024)

- United States of America* (52,375,047)
- Germany (16,750,084)
- Saudi Arabia (13,683,841)
- United Kingdom* (11,845,479)
- France* (9,186,757)

4.2 Top Origins (2024)

- India (18,533,845)
- China (11,701,619)
- Mexico (11,596,529)
- Ukraine* (9,769,216)
- Russian Federation (9,134,094)

4.3 Top Corridors (2024)

- Mexico → United States of America* (11,279,561)
- Afghanistan → Iran (3,752,317)
- Syrian Arab Republic → Türkiye (3,563,983)
- Russian Federation → Ukraine* (3,375,095)
- India → United Arab Emirates (3,248,545)

Chapter 5

Methodology

5.1 Network Model

Migration is modeled as a directed weighted graph $G = (V, E)$, where nodes are countries and edge weights represent migration stock.

5.2 Pipeline Flow Diagram

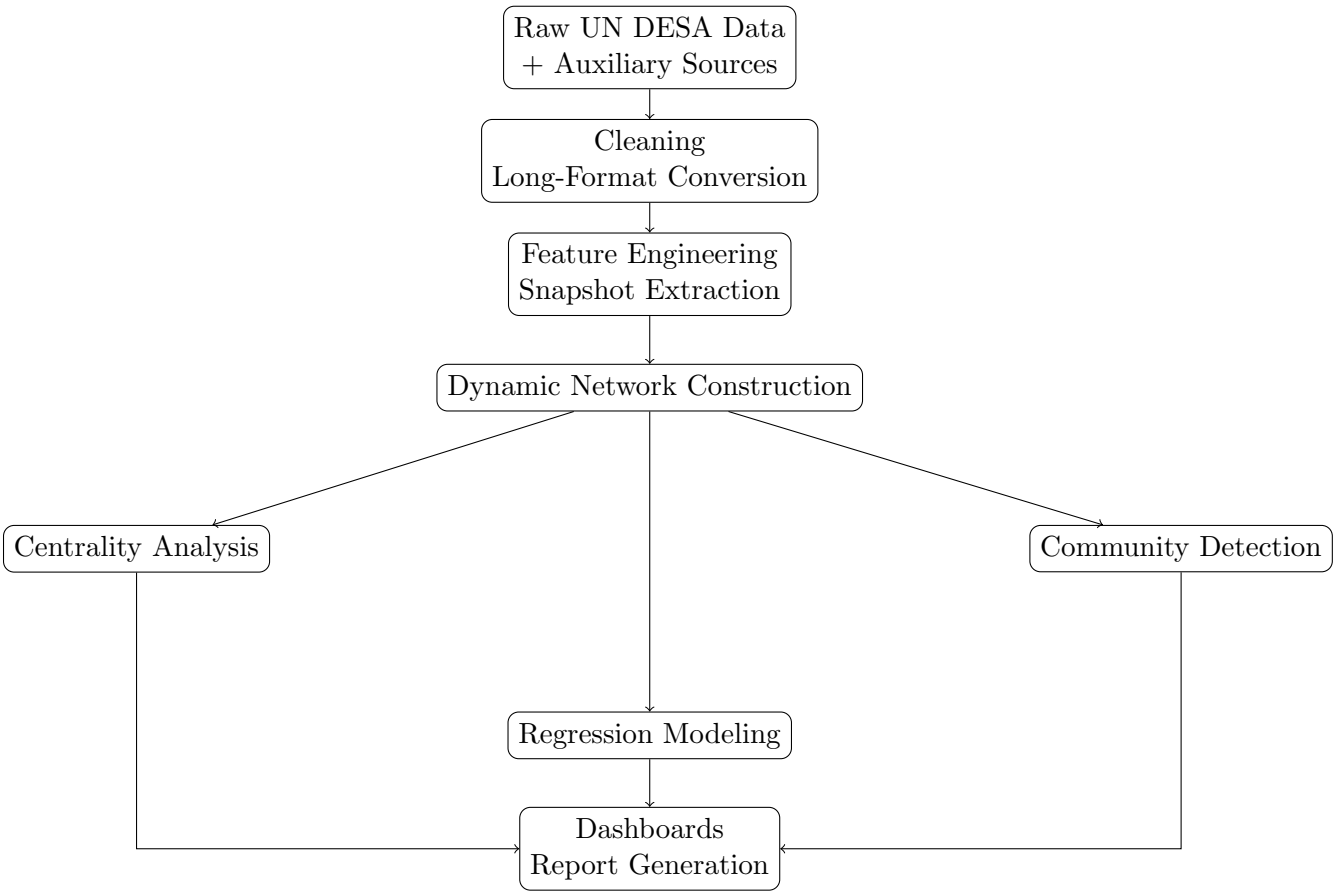

Figure 5.1: End-to-end analytical pipeline.

5.3 Centrality Metrics

We compute in-degree, out-degree, betweenness, closeness, and PageRank for each snapshot to identify hubs, bridges, and influential destinations.

5.4 Community Detection and Stability

Communities are identified using Louvain, Leiden, and Girvan–Newman. Stability is measured using NMI, ARI, and Jaccard drift between consecutive snapshots.

5.5 Custom Metric: Migration Influence Score (MIS)

$$\text{MIS}(\text{country}) = \text{PageRank}(\text{country}) \times \text{Total Migration Flow}(\text{country})$$

MIS highlights countries that are both structurally influential and high-volume destinations/origins.

5.6 Regression Modeling

We fit Linear, Ridge, Lasso, and Random Forest models with lagged flow features and origin/destination history. The best model achieves $R^2 \approx 0.95$.

Chapter 6

Results and Discussion

6.1 Key Findings

- The network is scale-free; a few hubs dominate global migration.
- Community structures are stable in developed regions but volatile in conflict-affected areas.
- Migration hubs persist across decades, indicating structural resilience.
- Temporal flows are strong predictors of future migration.

Chapter 7

Centrality Analysis (Visuals and Interpretation)

7.1 Degree Centrality

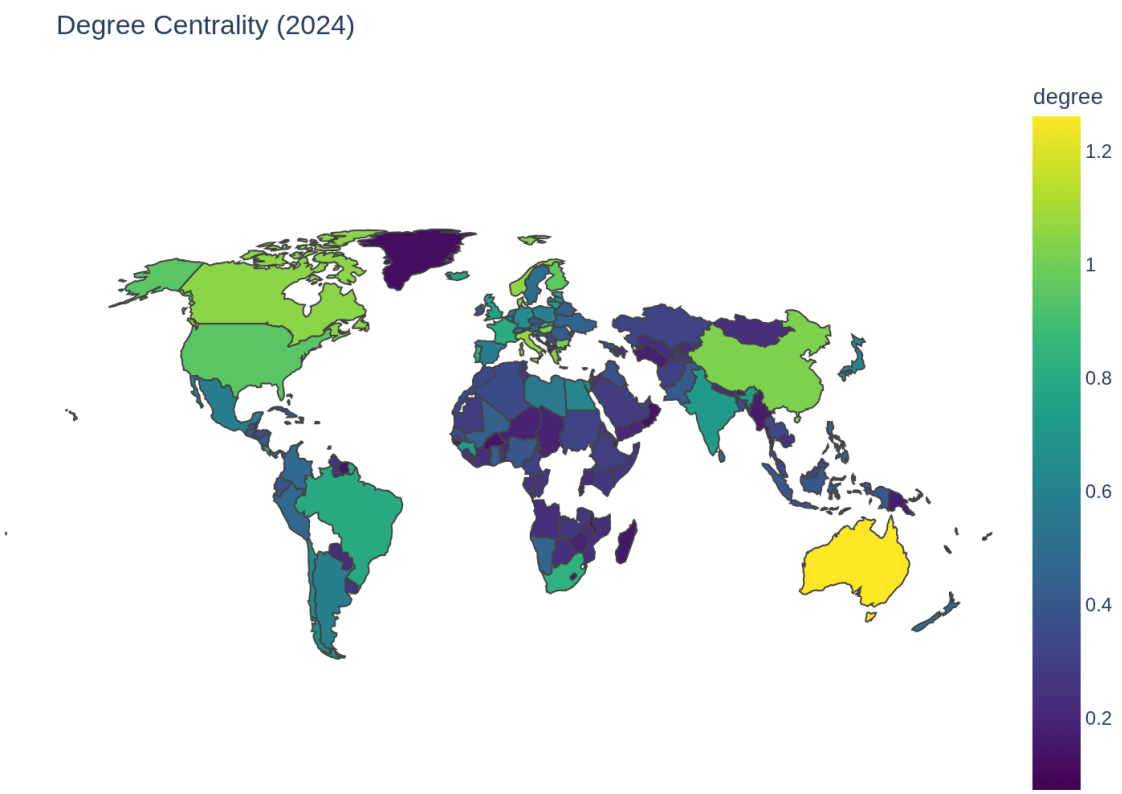

Figure 7.1: Degree centrality map (2024). Countries with broad bilateral connectivity are highlighted.

Explanation: Degree centrality measures the number of direct migration connections. High values indicate countries connected to many origins or destinations, reflecting breadth of links rather than depth of flows. In migration context, this highlights countries with diversified corridors and extensive global ties.

7.2 Betweenness Centrality

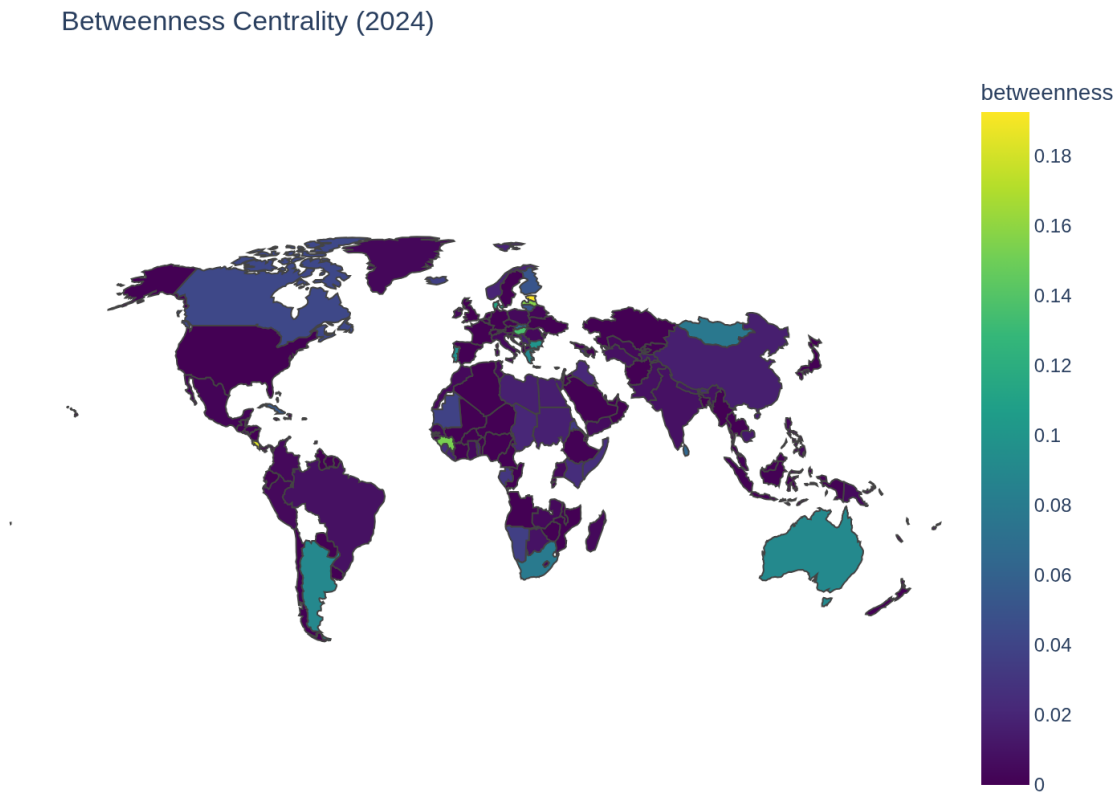

Figure 7.2: Betweenness centrality map (2024). Bridge countries between regions stand out.

Explanation: Betweenness captures how often a country lies on shortest paths between other countries. High values indicate transit or bridge roles that connect regional blocs. These countries are strategically important because disruptions here can affect multiple corridors.

7.3 Closeness Centrality

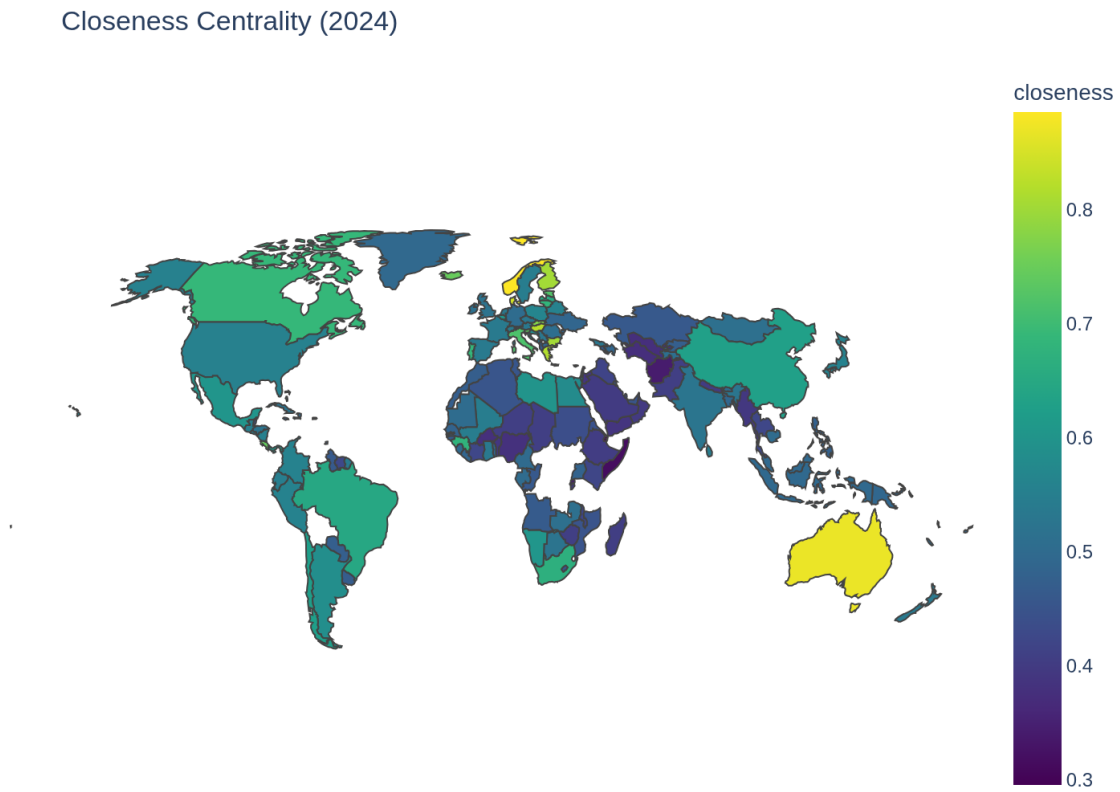

Figure 7.3: Closeness centrality map (2024). Centrally positioned countries appear with higher values.

Explanation: Closeness centrality measures how quickly a country can reach all others in the network. In migration terms, high closeness suggests structural accessibility and efficient reachability from many regions. This indicates a globally central position even when flows are moderate.

7.4 PageRank Centrality

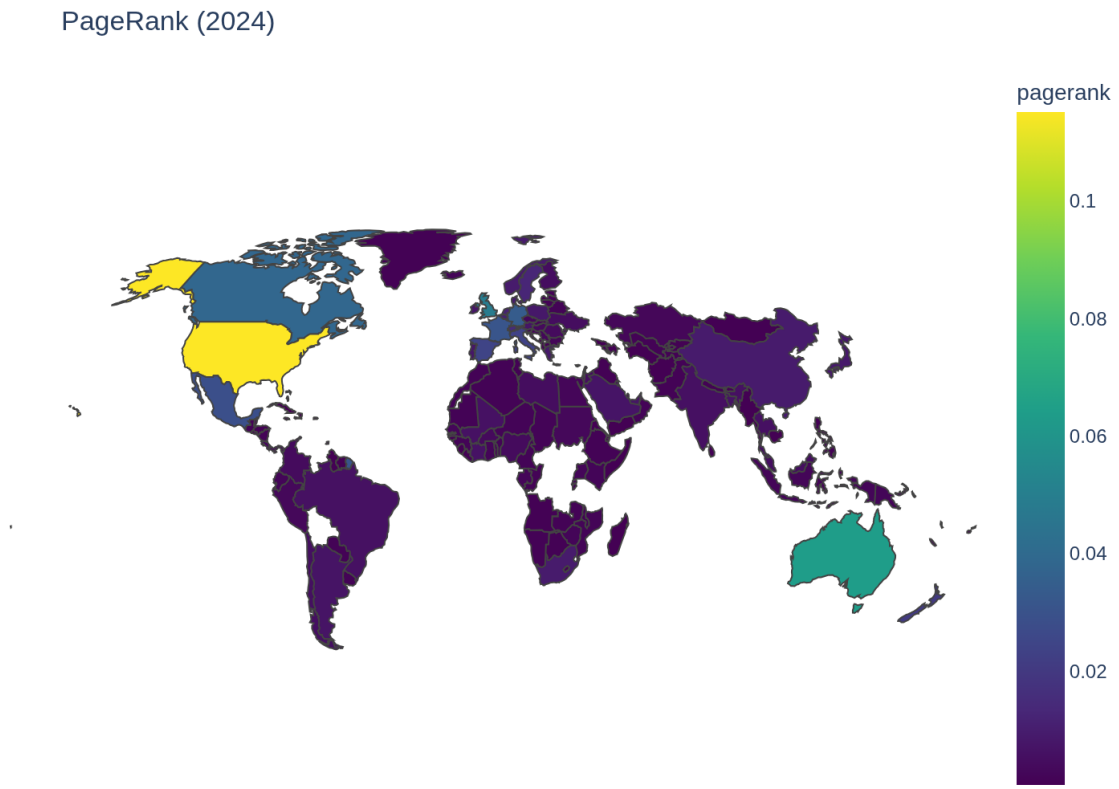

Figure 7.4: PageRank map (2024). Globally influential migration hubs are emphasized.

Explanation: PageRank rewards links from other influential countries, capturing global influence instead of simple connectivity. High PageRank countries are durable hubs with strong indirect influence. This differentiates major global destinations from locally connected nodes.

7.5 Network Visualization (Community Structure)

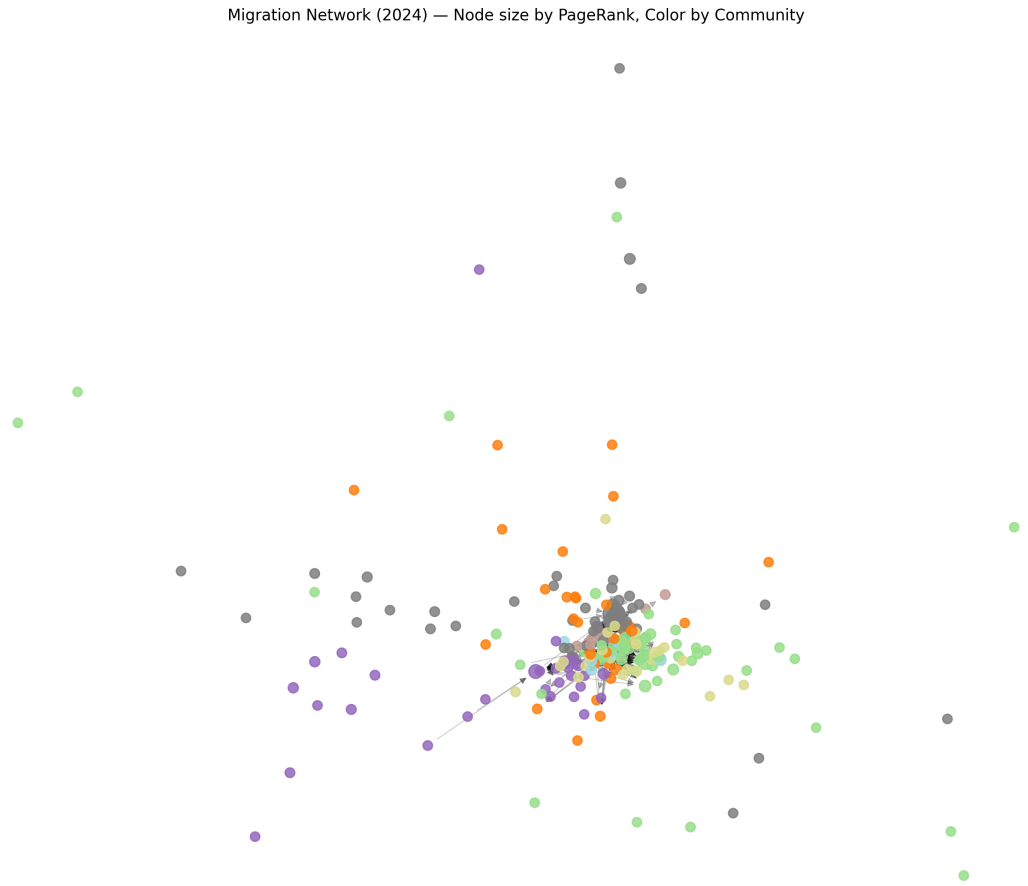

Figure 7.5: Migration network (2024) with node size proportional to PageRank and node color by community.

Explanation: Node size highlights global influence, while colors show community membership. The visualization reveals clustered regional blocs and a few dominant hubs, consistent with scale-free structure and persistent migration cores.

Chapter 8

Case Study: Syria → Türkiye

This corridor illustrates conflict-driven migration and shifting community membership. Türkiye acts as a regional bridge with high betweenness, linking origin regions to European destinations.

Chapter 9

Visual Outputs

9.1 5-Year Interval Visualizations

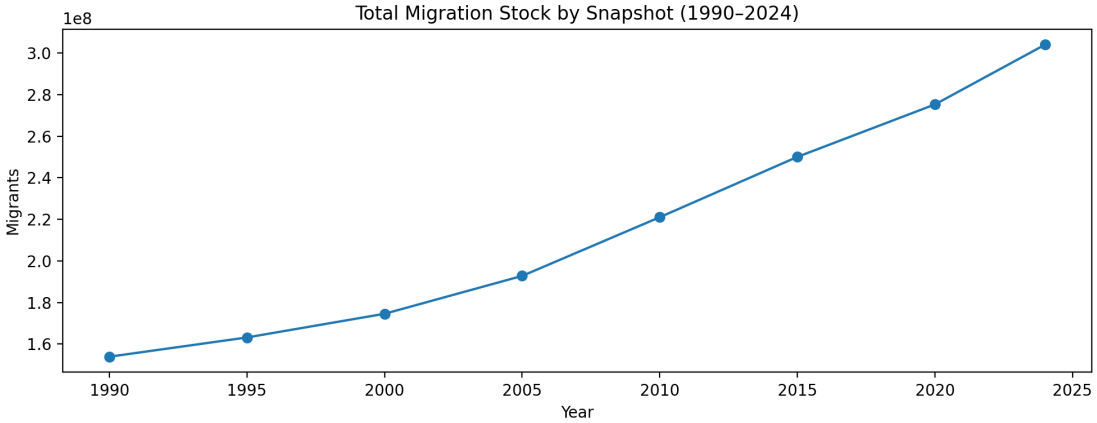

Figure 9.1: Total migration stock across snapshots (1990–2024).

Explanation: This trend line captures the long-run growth in global migration stocks. The steady rise indicates structural persistence and cumulative migration effects over decades.

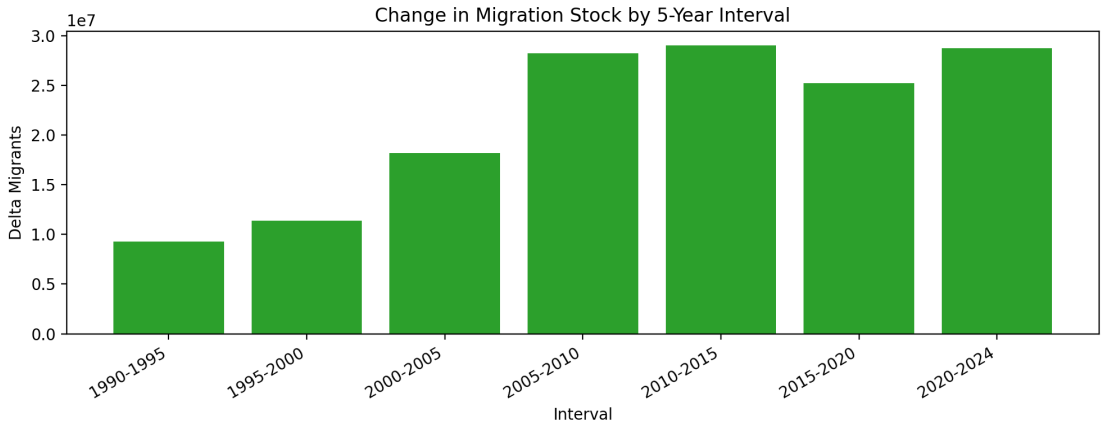

Figure 9.2: Change in migration stock by 5-year interval.

Explanation: Each bar shows the absolute change between consecutive snapshots (1990–1995, 1995–2000, . . . , 2020–2024). Larger increases indicate periods of accelerated migration growth.

9.2 Top Corridors per Interval

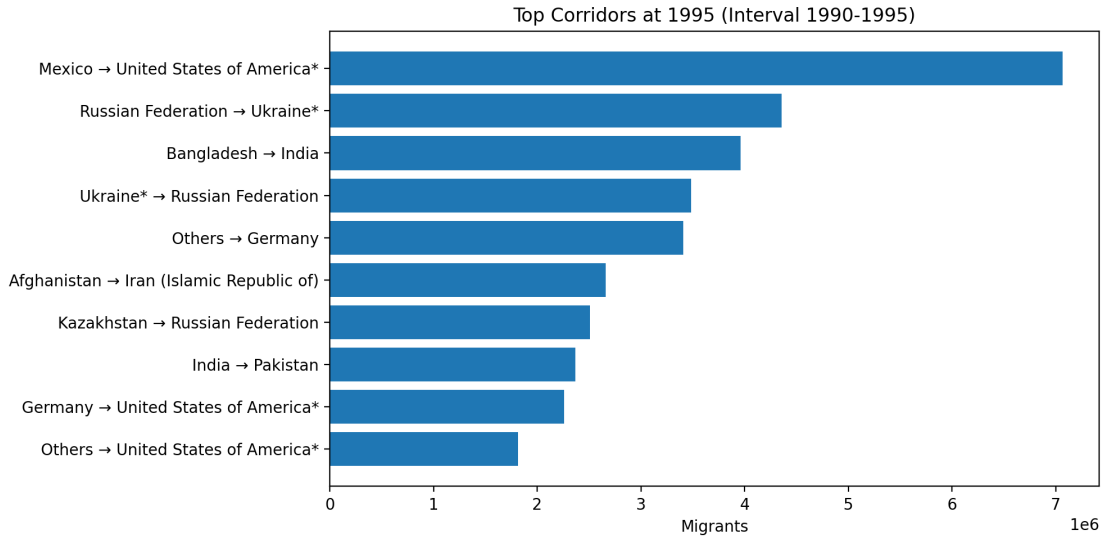

Figure 9.3: Top corridors at 1995 (interval 1990–1995).

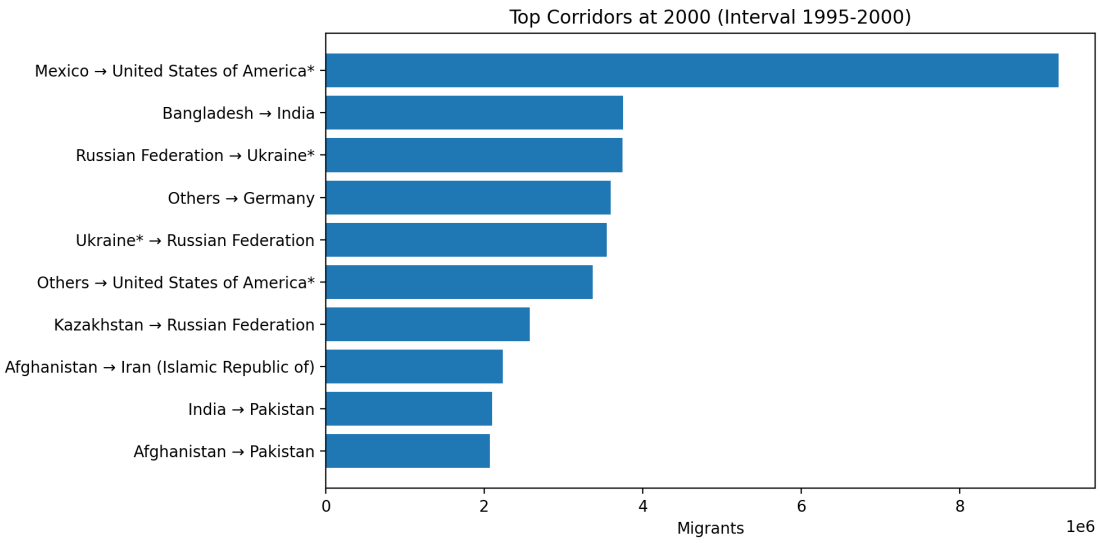

Figure 9.4: Top corridors at 2000 (interval 1995–2000).

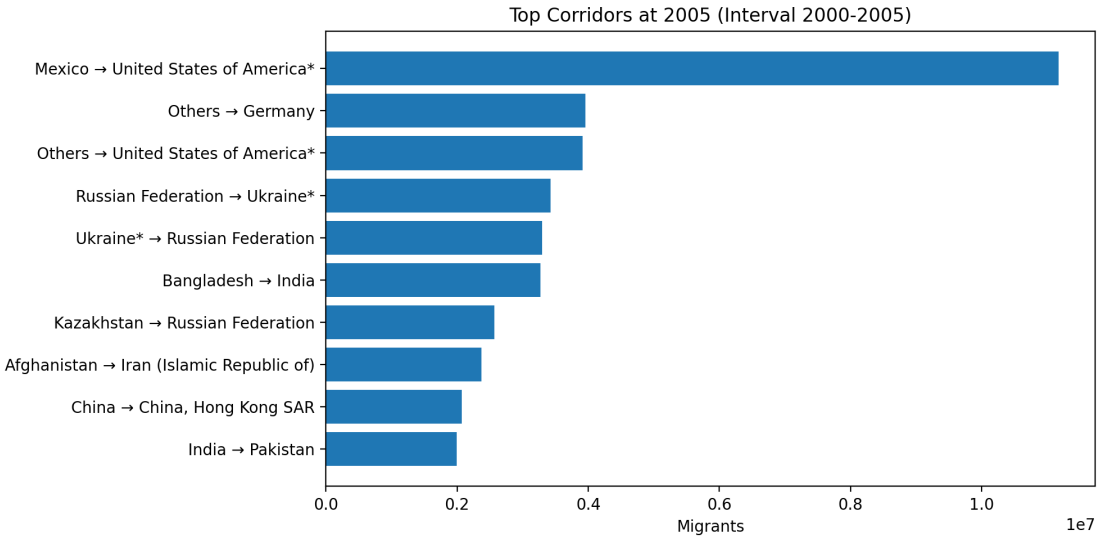

Figure 9.5: Top corridors at 2005 (interval 2000–2005).

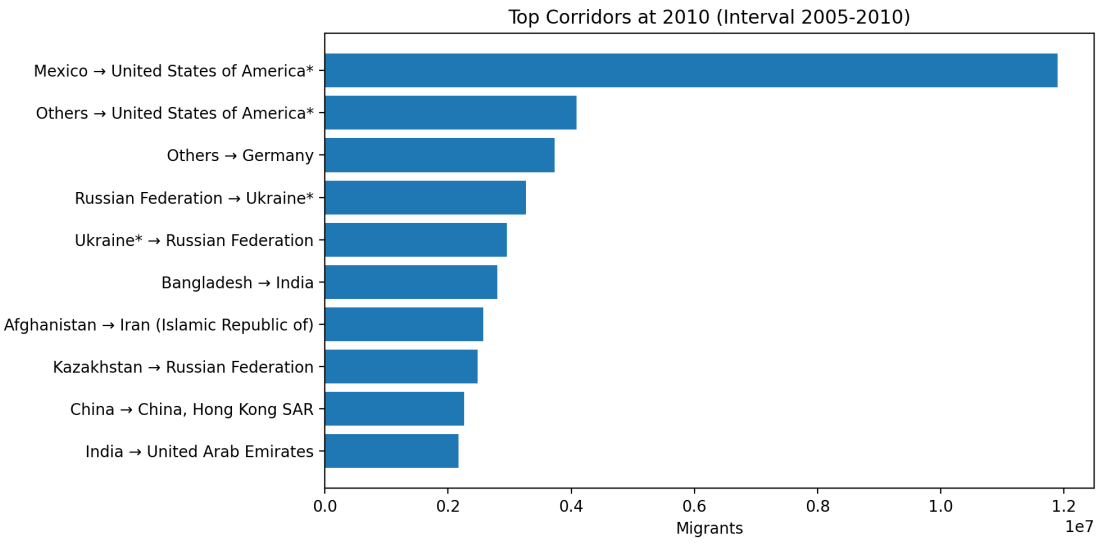

Figure 9.6: Top corridors at 2010 (interval 2005–2010).

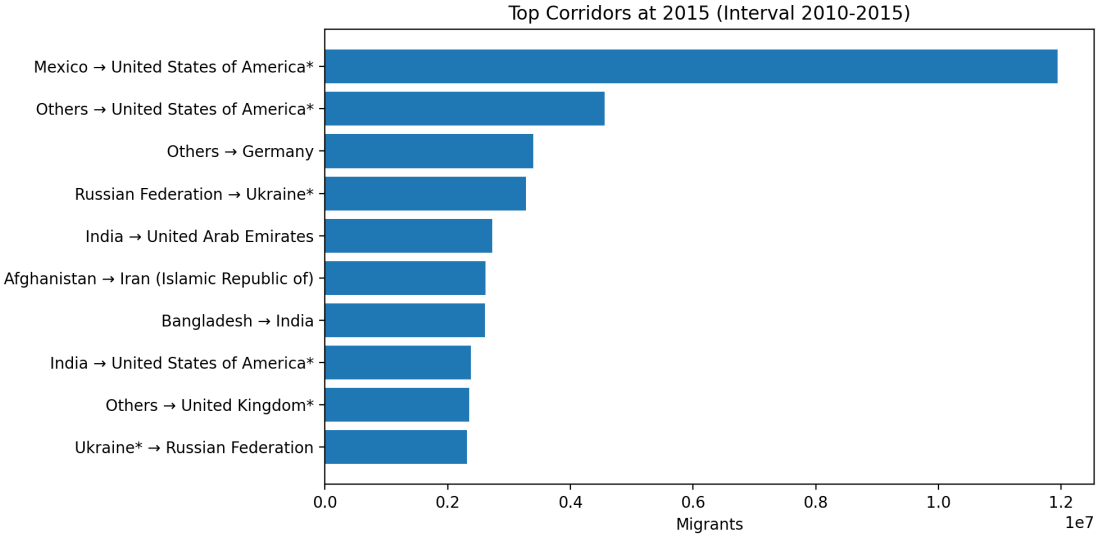

Figure 9.7: Top corridors at 2015 (interval 2010–2015).

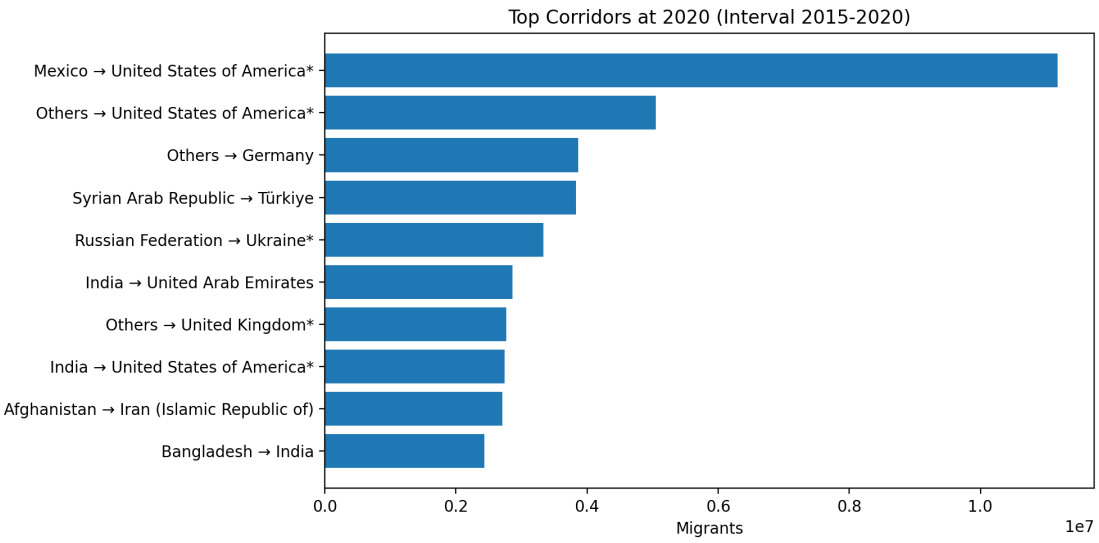

Figure 9.8: Top corridors at 2020 (interval 2015–2020).

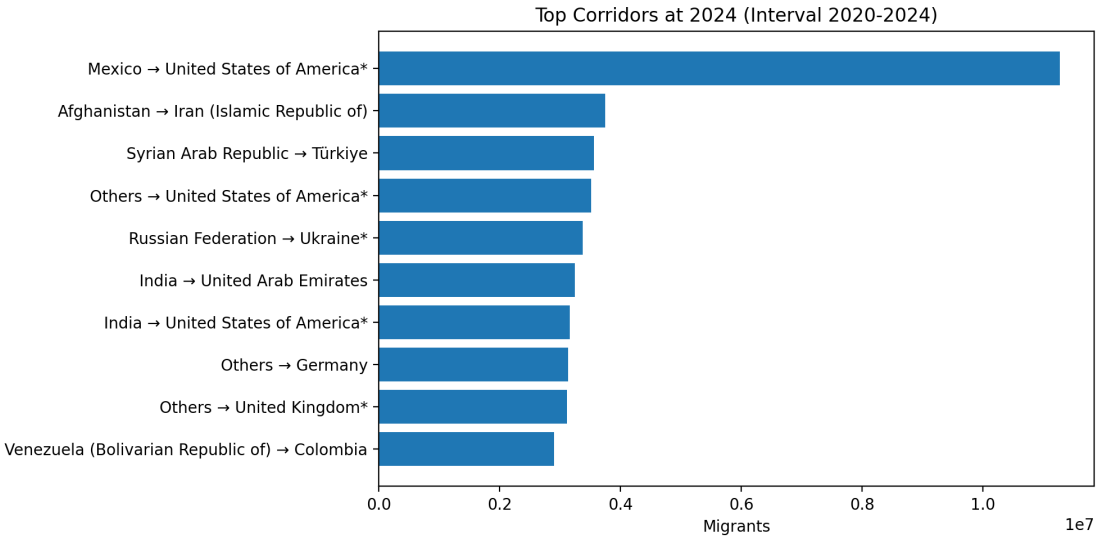

Figure 9.9: Top corridors at 2024 (interval 2020–2024).

Explanation: These panels show how leading corridors evolve by interval, highlighting persistent routes and emerging migration pathways.

9.3 Final Collage (All Visuals Summary)

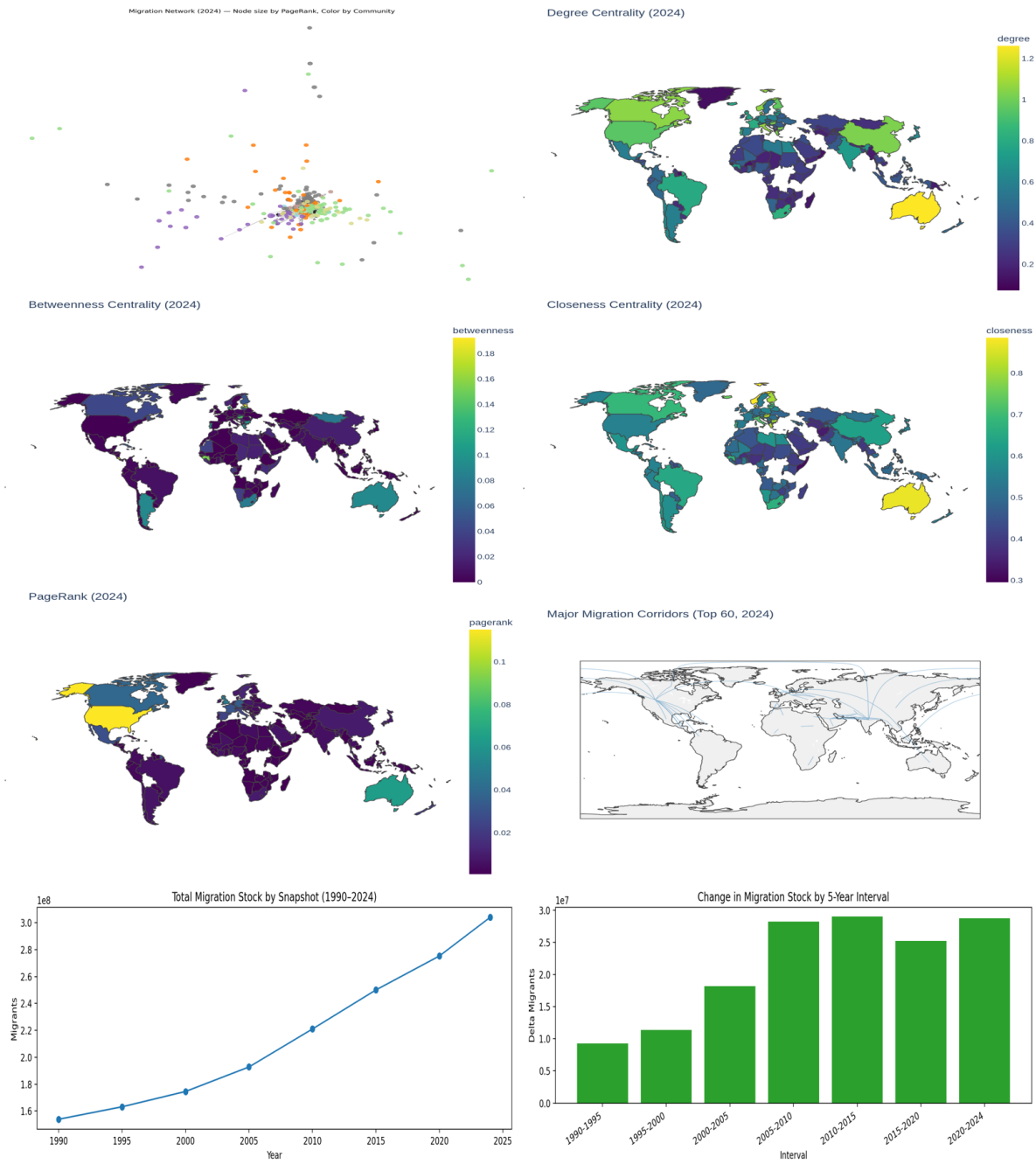

Figure 9.10: Collage summary of key network, map, and interval visuals.

Explanation: The collage provides a single-page visual overview of the project's strongest plots, suitable for presentation and quick review.

9.4 Major Corridors Map

Figure 9.11: Major migration corridors map (Top 60, 2024).

Explanation: The most intense corridors reveal dominant origin–destination pathways. These flows reflect geopolitical, economic, and conflict-driven factors. The visualization shows strong regional concentration with a limited set of global super-corridors.

9.5 Dashboard Screenshots

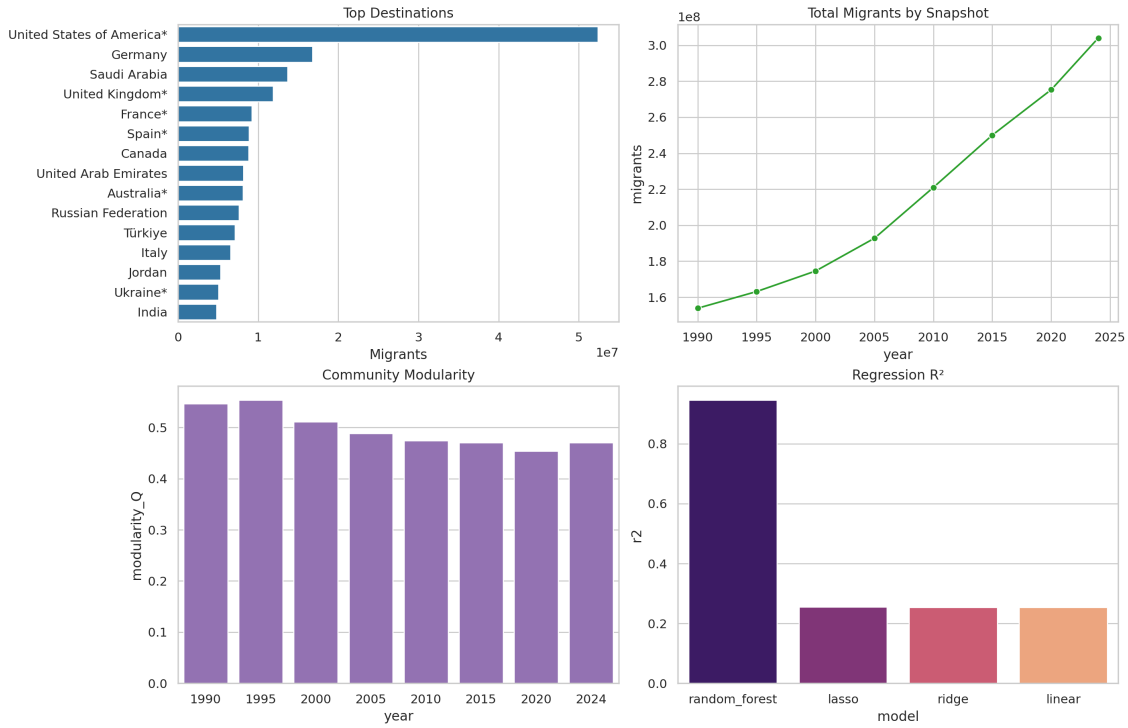

Figure 9.12: Executive Overview Dashboard

Explanation: Summarizes total migration volume, top destinations, modularity trends, and model performance.

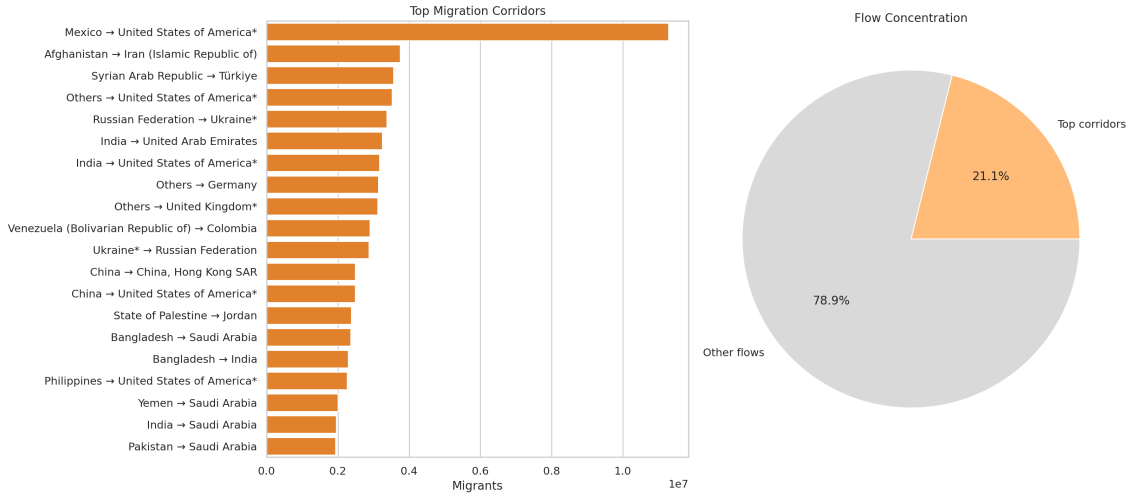

Figure 9.13: Top Migration Corridors Dashboard

Explanation: Highlights the strongest bilateral flows and their distribution.

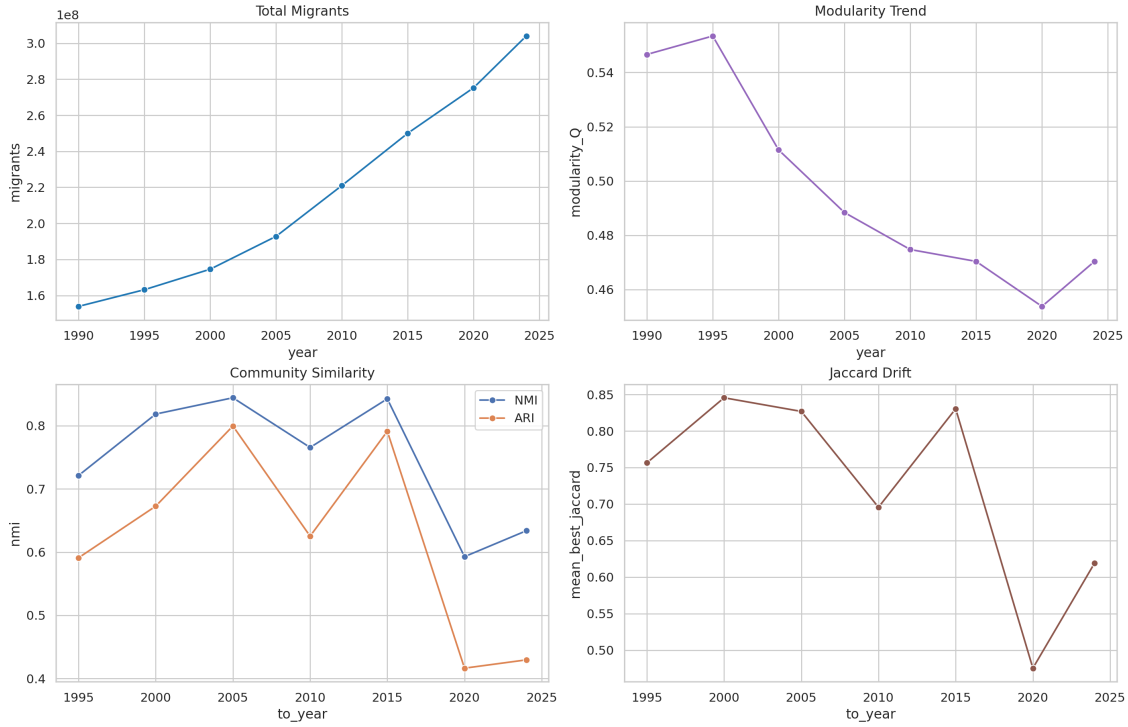

Figure 9.14: Temporal Trends Dashboard

Explanation: Tracks total migrants over time alongside stability indicators.

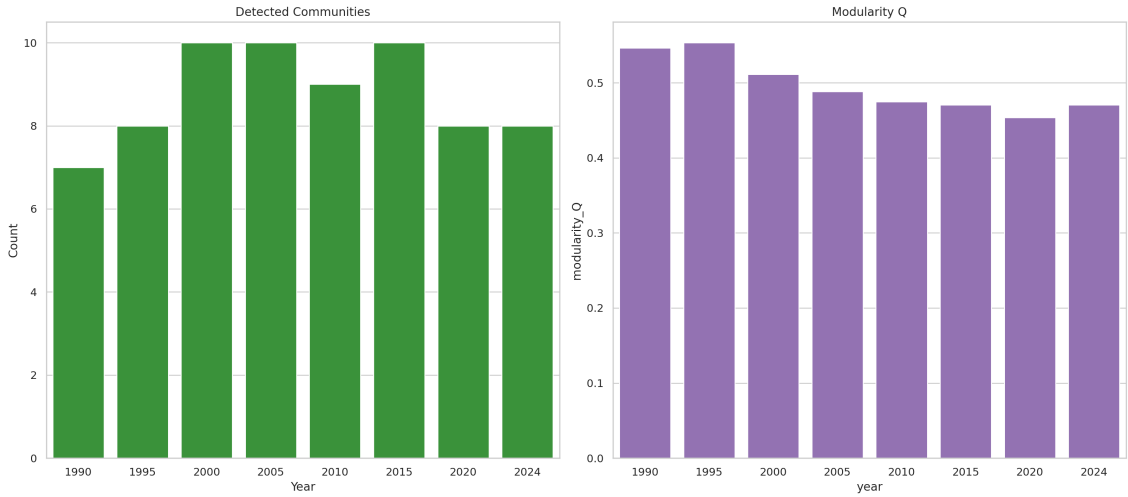

Figure 9.15: Community Structure Dashboard

Explanation: Displays the number of detected communities and modularity over time.

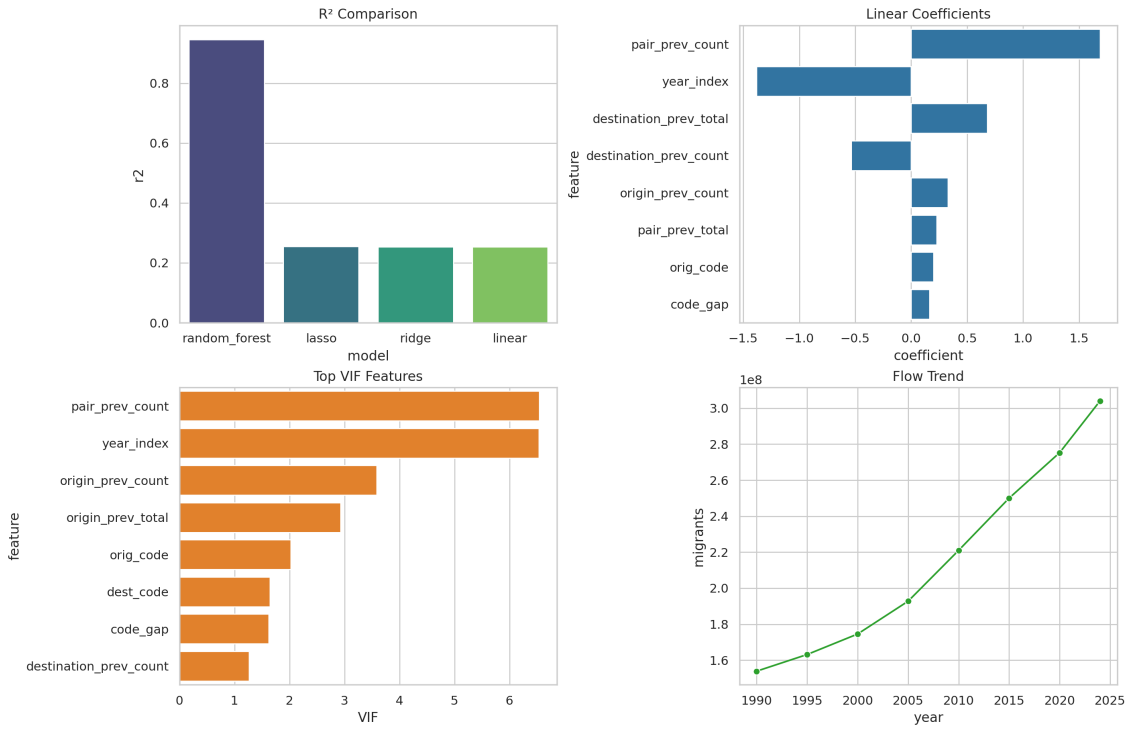

Figure 9.16: Regression Diagnostics Dashboard

Explanation: Shows model performance, coefficient patterns, and multicollinearity checks.

Chapter 10

Limitations and Future Work

10.1 Limitations

- Informal or undocumented migration is not fully captured.
- Auxiliary indicators are incomplete across countries and years.
- Community results are sensitive to algorithm choice.
- Policy shocks are not explicitly modeled.

10.2 Future Work

- Integrate real-time migration APIs and administrative data.
- Apply Graph Neural Networks (GNNs) for prediction.
- Add climate and policy variables for scenario modeling.
- Extend forecasting to multi-scenario simulations.

Chapter 11

Conclusion

This project demonstrates that combining network science with predictive modeling produces actionable migration insights. The centrality visualizations highlight persistent hubs, while the 5-year interval analysis reveals how migration intensifies across decades. Community stability metrics and the custom MIS score provide original insights into structural influence beyond raw volume. The final collage and dashboards deliver a clear, presentation-ready summary of findings for academic or policy audiences.

Chapter 12

References

- United Nations, Department of Economic and Social Affairs. *International Migrant Stock 2024*.
- Newman, M. E. J. (2010). *Networks: An Introduction*. Oxford University Press.
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks.
- Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: guaranteeing well-connected communities.

Appendix A

Appendix A: Algorithm Summary

Algorithm	Description
Louvain	Modular community detection optimizing modularity.
Leiden	Improved modularity algorithm with well-connected communities.
Girvan–Newman	Edge-betweenness-based hierarchical community detection.
PageRank	Random-walk-based influence score.
Betweenness	Frequency of node on shortest paths.

Appendix B

Appendix B: Regression Features

- Pair historical totals and counts
- Origin and destination historical totals
- Year index and code gap
- Lag-based predictive features

Appendix C

Appendix C: Output Artifacts

- Centrality tables per snapshot
- Community partitions and modularity tables
- Jaccard drift chart and NMI/ARI comparison tables
- Regression coefficient and VIF reports
- Dashboard-ready CSVs and screenshots